Matching Division Stories, Drawings, and Equations

Matching division stories to equations, reading equal-group and array drawings, choosing the story an equation tells, and checking with multiplication fact families

The one idea behind every problem. A division sentence always says the same three things: **how many in all**, **how many groups**, and **how many in each group**. A story, a drawing, and an equation are three ways of telling that one idea. Your job is to move between them.

total in all	÷	number of groups	=	how many in each group
total in all	÷	how many in each group	=	number of groups

Reference card. No numbers are filled in — use the numbers given in each problem.

Directions.

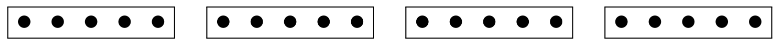
- Every problem asks you to show the match, not just the answer: write the division equation, or draw the groups, or say which story fits.
- You may draw dots, circles, or an array in the work space — pictures are part of the work here.
- Write your final answer on the answer line.

1. Maya has 12 stickers. She shares them equally with 3 friends at her table, so that each friend gets the same number.

Write the division equation this story tells, then answer it: how many stickers does each friend get?

Answer: ____________________

2. The picture below shows 20 counters that have already been sorted into 4 equal groups.



Write the division equation that matches the picture, then answer it: how many counters are in each group?

Answer: ________________

3. Here is a multiplication sentence with a missing number:

$$3 \times \square \;=\; 18$$

(a) What number belongs in the box?

(b) Write the division sentence that belongs to the same fact family and asks for that same missing number.

Answer: ________________

4. A teacher has 24 crayons. She packs them into boxes, and each box holds 6 crayons. This story does *not* tell you the number of groups — it tells you how many go in each group. Write the division equation the story tells, then answer it: how many boxes does she fill?

Answer: ________________

5. Look at this equation:

$$15 \div 5 = \square$$

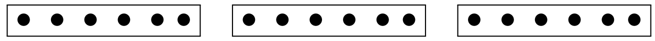
Which story below matches the equation? Circle A or B, tell in one sentence how you know, then answer the equation.

A. There are 15 shells. They are put into 5 equal piles.

B. There are 15 shells in each of 5 buckets.

Answer: ________________

6. The picture below shows 18 counters sorted into 3 equal groups.



Write the division equation that matches the picture, then answer it. Then write the multiplication equation that matches the same picture.

Answer: ____________

7. A fact family uses the same three numbers over and over.

(a) Fill in the missing number: $7 \times \square = 28$

(b) Write all four sentences of this fact family — two multiplication sentences and two division sentences.

Put the missing number from part (a) on the answer line.

Answer: ____________

8. Read carefully — one of these numbers is the total and one is the size of each group. A farm stand has 32 apples. The apples are put into baskets with 8 apples in each basket. Write the division equation this story tells, then answer it: how many baskets are used?

Answer: ____________________

9. Look at this equation:

$$42 \div 6 = \square$$

(a) Only one of these stories matches the equation. Circle it.

A. A baker makes 42 muffins and puts 6 muffins in each tray.

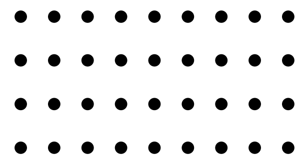
B. A baker makes 6 trays of muffins with 42 muffins on each tray.

(b) Explain in one sentence why the other story does not match.

(c) Answer the equation and write it on the answer line.

Answer: ____________________

10. The array below shows 36 counters arranged in 4 equal rows.



Write the division equation that matches the array, then answer it: how many counters are in each row? Be careful — the question asks about a row, not about the number of rows.

Answer: ____________________

11. This time the missing number is the one you divide *by*:

$$45 \div \square \;=\; 9$$

(a) What number belongs in the box?

(b) Write a short story that this equation could tell, using your answer from part (a).

Put the missing number from part (a) on the answer line.

Answer: ____________________

12. Challenge. A classroom has 5 packs of pencils, and every pack holds 6 pencils. All of the pencils are shared equally among 3 tables.

(a) What do you have to find *before* you can divide? Write that step as an equation.

(b) Now write the division equation and answer it: how many pencils does each table get?

(c) Draw a picture that shows your division. Label what the groups are.

Answer: ____________________